Performance optimisation

Outline

- Occupancy
- Branching
- Memory optimisations
- Miscellaneous

We will use the AMD MI210 GPU for quantitative examples

Other GPUs have similar designs, but the numbers will differ!

Occupancy

Filling the GPU

- GPU devices are high latency, high bandwidth devices
- You need to have enough tasks to fill all memory/compute units on the devices
- You need to have enough tasks to exploit the high bandwidth and hide latency

AMD MI210

- 104 CU (Compute units)
- Each CU has 4 SIMD processors with 16 lanes
- Up to 8 wavefronts can be executed by a SIMD processor
- A wavefront is composed of 64 threads
- All the threads in a wavefront execute the same instruction

Occupancy

- Maximum number of threads on the GPU= 104 (CU) x 4 (SIMD processors) x 8 (Instructions per SIMD) x (64 threads per wavefront) = 212992
- Over 200k threads can be scheduled on the GPU
- The ratio between the number of threads executing on a GPU and the theoretical maximum number of threads scheduled on the GPU is called occupancy

Occupancy

- Some of the wavefronts on the device will be stalled, i.e. waiting for data to become available from memory, floating point operation units being busy …
- But the GPU can quickly switch between wavefronts that are stalled and wavefronts that are ready to execute, hiding latency
- Need occupancy large enough to hide latency
- Higher occupancy does not always equal higher performance.

Occupancy limiters

Theoretical maximum occupancy can be limited by

- Team size and number of teams
- Shared memory
- Registers

Occupancy: Teams size and number

- Each team is scheduled to a single CU
- Need at least as many teams as Computing Units to fill the GPU
- Up to 32 wavefronts (2048 threads) can be scheduled per CU
- If the team size is 17 wavefronts (1088 threads), only one team can fit on a CU at the same time. Max. possible occupancy is 1088/2048 = 53%

Occupancy: Teams size and number

- Use the **`num_teams`** clause to specify the number of teams
- Use **`thread_lim`** to place an upper bound on the number of threads per team
  - cannot specify an exact number as some devices cannot support this

Occupancy: Teams size and number

- Make sure there is enough parallelism to fill the GPU
- Merge small loops into bigger loops
- In multidimensional loop nests use the `collapse` clause to spread inner loop operations across different threads

Occupancy: Register pressure

- Each team needs a certain number of registers
- Finite number of registers per computing units and SIMD, hardware dependent
- This can limit the number of teams that are executed concurrently on Computing Unit
- If more registers are needed than are available, the compiler can decide to spill the registers onto shared or global memory.
- Register memory is orders of magnitude faster than shared or global memory

Occupancy: Register pressure

- MI210 has **Vector Registers** (**VGPRs**) and **Scalar Registers** (**SGPRs**)

- A vector register holds 64 items, 4B wide
- Up to 512 vector registers per SIMD processor

- A scalar register holds the same value for all threads in a wavefront.
- Up to 800 per SIMD
- In practice, it is more common to run out of vector registers

Occupancy: Register pressure

- If a wavefront needs 128 vector registers, at most 512/128 = 4 wavefronts can reside on a SIMD, and so only 16 out of a possible 32 wavefronts can run on a CU.
- Occupancy is 50%

- N.B. a single wavefront cannot use more than 256 registers

Occupancy: Register pressure

- Split big loop bodies into multiple loops with smaller bodies. Smaller loop bodies might need smaller number of registers
- Smaller loops might decrease the amount of parallelism, so might not improve performance.
- May require additional temporary arrays to transmit values between loops
  - increases memory usage, can cause more pressure on cache space and/or memory bandwidth
- Use the `thread_limit` clause. If the compiler knows at compile time the maximum number of threads per team it can allocate registers more efficiently.

Occupancy: Shared Memory

- OpenMP tends to not to utilise shared memory very much.
- Shared memory is shared by all threads in a team.
- On MI210, there is 64KB of shared memory per CU.
- If a team requires 32KB of shared memory , at most two teams can fit on a single Computing Unit

Branching

Branching

- Threads in a team are grouped in wavefronts / warps
- The number of threads in a wavefront is architecture dependent
- For AMD GPUs a wavefront is usually 64 threads
    - For NVIDIA GPUs a warp is usually 32 threads
- All threads in a wavefront always execute the same instruction

Branching

- What if different threads in a wavefront execute different instructions ?

```c
#pragma omp target teams
distribute parallel for
for ( int i=0; i<n; i++){
    if (i%2 == 0) {
        x[i] = a
    }
   else {
        x[i] = b;
    }
}
```

C/C++

```fortran
!$omp target teams
        distribute parallel do
do i=1,n
        if (modulo(i,2) == 0) then
                x(i)= a
        else
                x(i)=b
        endif
end do
!$omp end target teams
```

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Branching

- Set x to a on all threads, but mask odd threads


```
if (i%2 == 0)
{
    x[i] = a
}
```

C/C++

- Set x to b on all threads, but mask even threads


```
else
  {
      x[i] = b;
  }
```

C/C++

- Need to run two instructions per thread instead of one instruction per thread

Branching

- For each instruction, run the instruction on all threads, but mask the result of threads that do not need to run the instruction
- Each branch executed by all threads: leads to loss of parallelism
- Can strongly limit the performance of some scientific codes.
- An example of codes strongly affected are Monte Carlo codes.
  - Monte Carlo codes often execute a random event per thread. All threads then execute different instructions at the same time, leading to serialization

Branching

- Avoid branching in GPU codes
- Might require re-thinking the parallel algorithm

Memory optimization

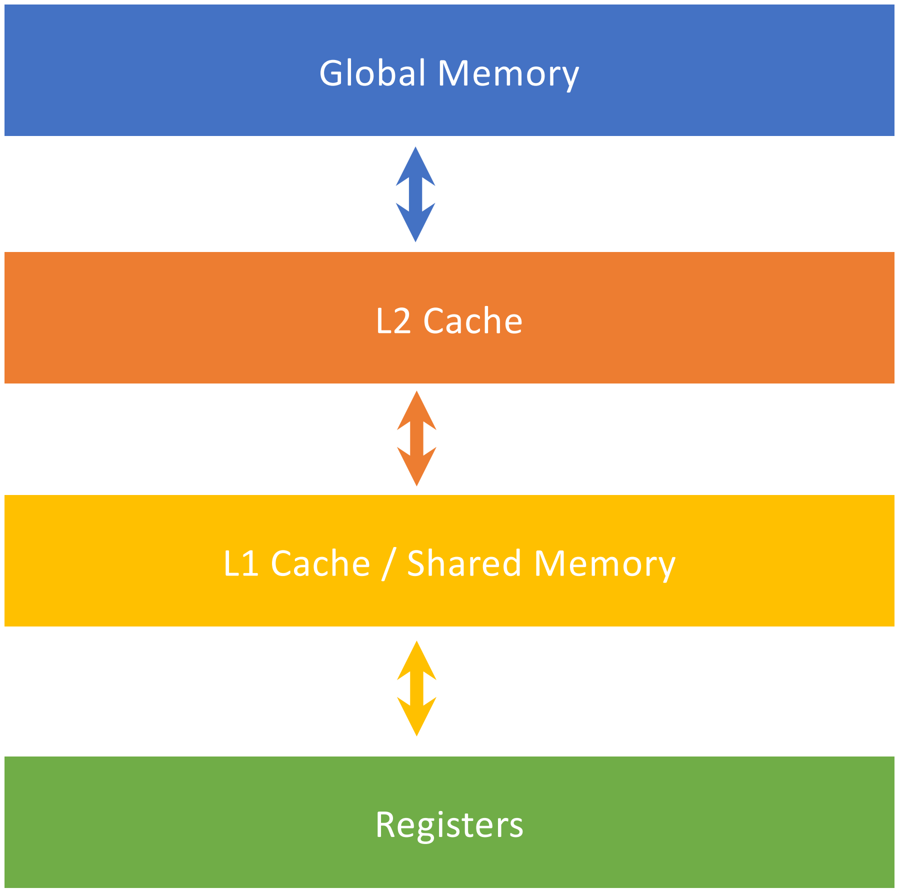
Memory usage

- Hierarchical memory structure
- Different types of Memory: L2 cache , L1 cache/Shared memory, Global Memory etc...
- Each layer is faster than the one above

Memory usage

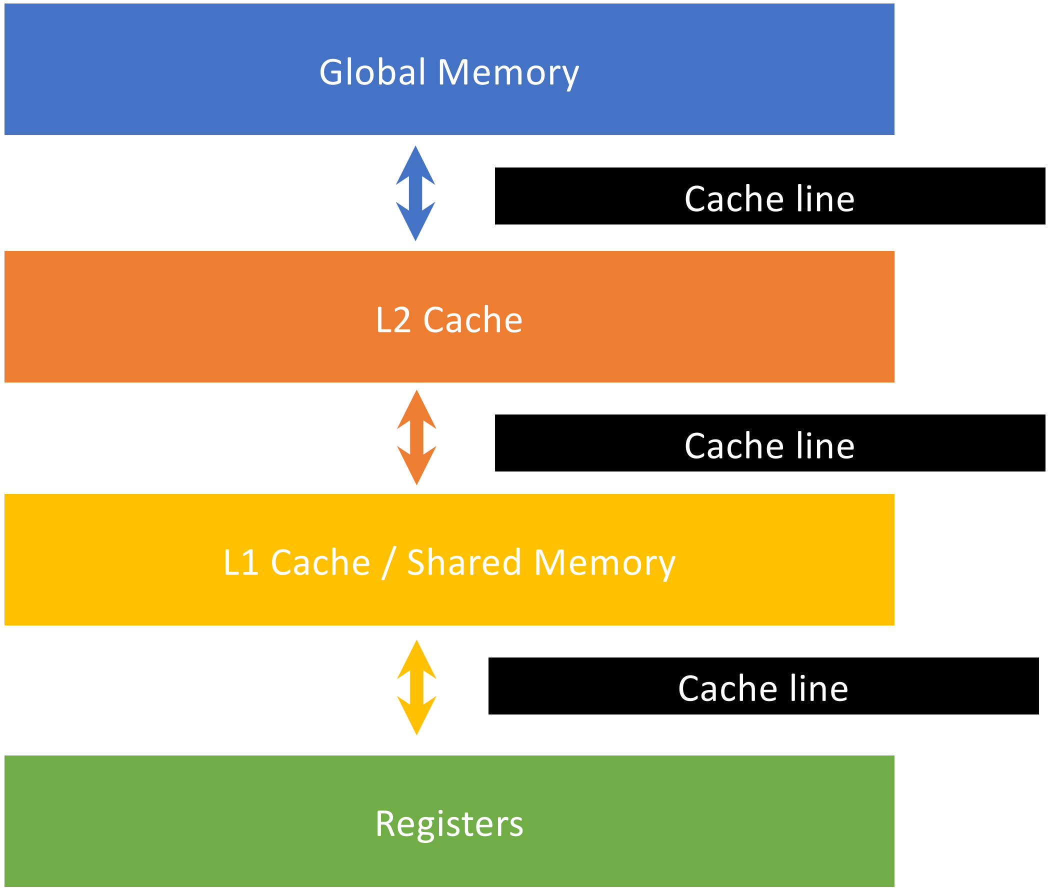
- Global Memory:
  - Visibility: all Compute Units
  - Capacity: ~ 100 GiB
- L2 cache :
  - Capacity: ~ 10 MiB per compute die
  - Visibility: a group of Compute Units on the same die
- L1 cache / shared memory :
  - Visibility: private to a Compute Unit
  - Capacity: ~ 60 KiB per Compute Unit

Memory usage



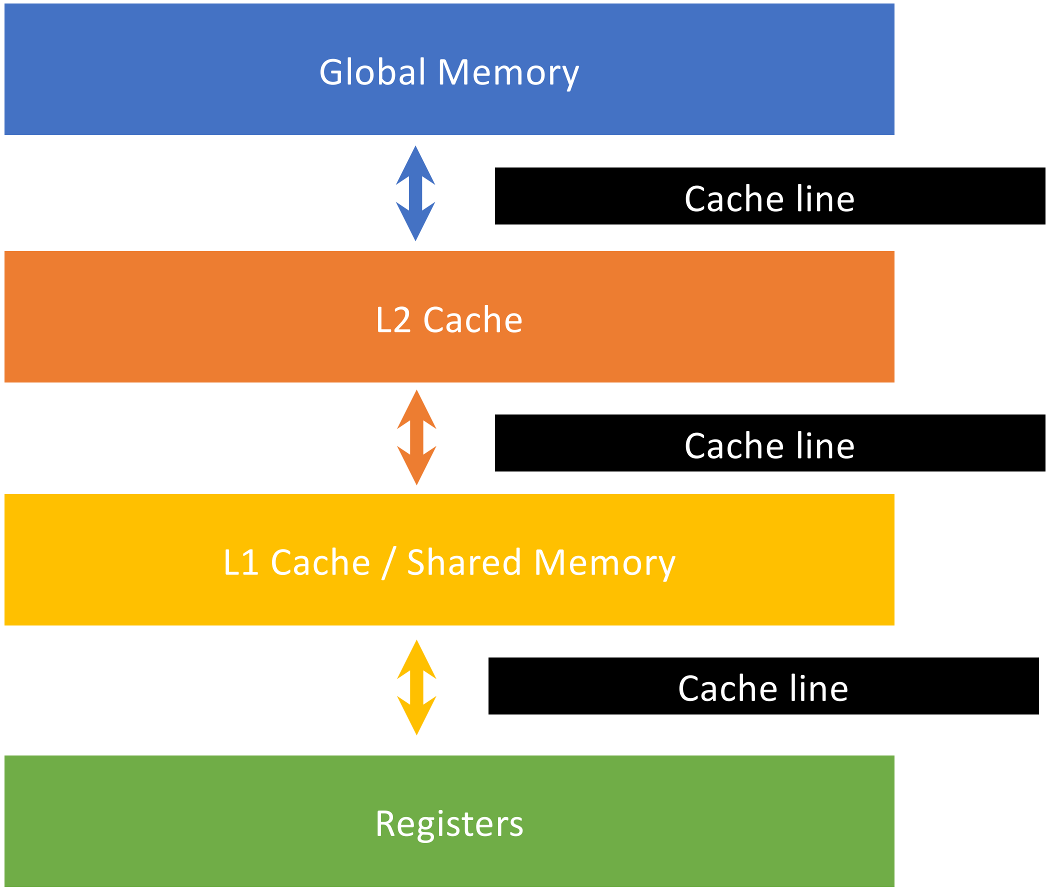
- All data needs to travel from global memory to registers across all layers
- If data is not found in a layer it needs to find the data in the upper layer
- As bottom layers have smaller capacity, some of the data might need to be overwritten

Memory usage



- All data is transferred in bundles of a fixed size, called a cache line
- Cache line size depends on hardware. Usually 64 byte (e.g. MI210) or 128 bytes (e.g. A100)
- If threads in the same wavefront access the same cache line, only one load instruction is needed: accesses are coalesced

Memory usage



- Aim to reduce the amount of data traffic between layers
- Try to reuse as much as possible data already in the layers (no different to normal CPUs!)

Coalesced memory access

```
#pragma omp target teams distribute parallel for
for (int i=0; i<n; i++){
  a[i]= b[i] + c;
}
```

C/ C++

```
!$omp target teams distribute parallel do
   do i=1,n
       a(i) = b(i) + c
   end do
!$omp end target teams
```

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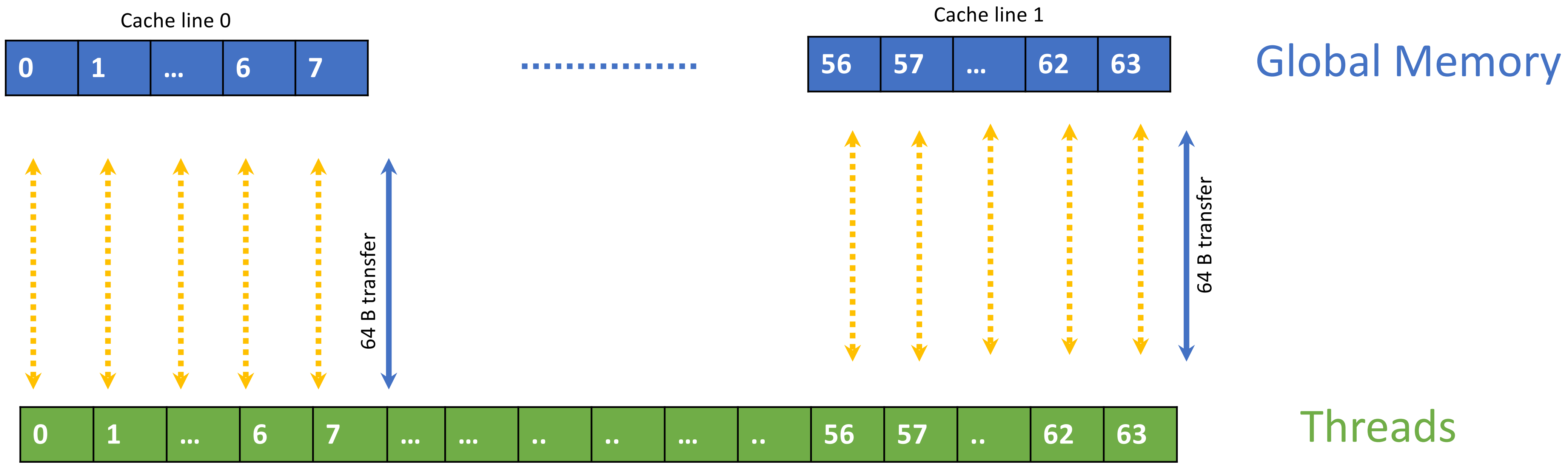
Coalesced memory access



- 64 double precisions elements loaded from b
- 8 LOAD instructions per wavefront from b
- 64B x 8 = 512B transferred from b

Coalesced memory access

```
#pragma omp target teams distribute parallel for
for (int i=0; i<n; i++)
{
    a[i]= b[i*8] + c;
}
```

C/C++

```
!$omp target teams distribute parallel do
    do i=1,n
        a(i) = b(i*8) + c
    end do
!$omp end target teams
```

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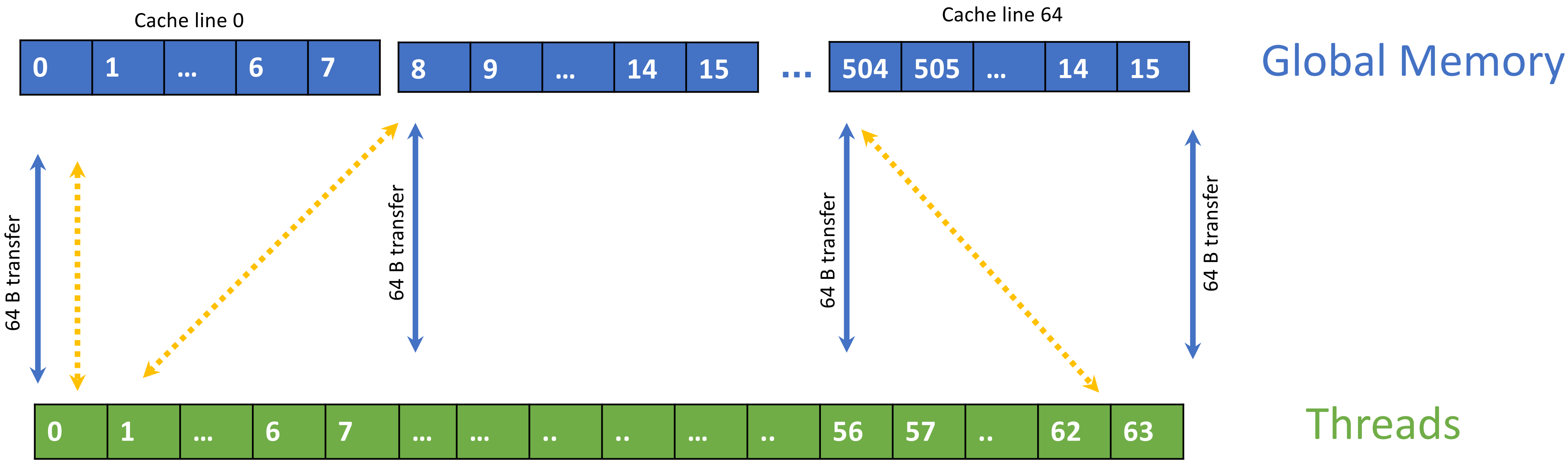
Coalesced memory access



- 64 double elements loaded from b
- 64 LOAD instructions per wavefront from b
- 64 B x 64 = 3.9 KiB transferred from b

Coalesced memory access
Loop ordering

```
#pragma omp target teams distribute\
parallel for collapse(2)
for ( int i= 0; i<n; i++) {
  for ( int j= 0; j<n; j++) {
      a[i][j]= b[i][j] + c;
  }
}
```
C/C++

```
!$omp target teams distribute parallel do
  do j=1,n
    do i=1,n
       a(i,j) = b(i,j) + c
    end do
  end do
!$omp end target teams
```
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- Neighbouring inner loop iterations are often assigned to neighbouring threads.

Coalesced memory access
Loop ordering

```
#pragma omp target teams distribute parallel for collapse(2)
for (int i= 0; i<n; i++) {
  for (int j= 0; j<n; j++) {
      a[i][j]= b[i][j] + c;
  }
}
```

C/C++

- In C/C++ rightmost array dimension is contiguous in memory
- j and j+1 element likely to be scheduled on the same wavefront
- Memory accesses are coalesced

Coalesced memory access
Loop ordering

```
#pragma omp target teams distribute parallel for collapse(2)
for (int j= 0; j<n; i++) {
  for (int i= 0; i<n; i++){
    a[i][j]= b[i][j] + c;
  }
}
```
C/C++

- Strided access between iterations with index i and i+1
- Memory accesses are not coalesced

Coalesced memory access
Loop ordering

```
!$omp target teams distribute parallel do collapse(2)
  do j=1,n
    do i=1,n
      a(i,j) = b(i,j) + c
    end do
  end do
!$omp end target teams
```

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- In Fortran, leftmost array dimension is contiguous in memory
- i and i+1 elements likely to be scheduled on the same wavefront
- Memory accesses are coalesced

Coalesced memory access
Loop ordering

```
!$omp target teams distribute parallel do collapse(2)
  do i=1,n
    do j=1,n
       a(i,j) = b(i,j) + c
    end do
  end do
!$omp end target teams
```

F

- Strided access between iterations with index j and j+1
- Memory accesses are not coalesced

Miscellaneous

Kernel latency

- Scheduling a team on the device takes time, regardless of how long it takes to execute
- Latency depends on size of the team, number of `firstprivate` variables used, etc.
- On MI210 a typical latency for offloading a computational kernel is about 5-10 microseconds

Kernel latency

- Try to have sufficiently long computation kernels, in order to hide kernel launch latency
- Merging small computational kernels into bigger computational kernels can help

Data movements

- Moving data between CPU and GPU is slow. Avoid un-necessary data transfers
- Use `enter data` / `exit data` to avoid transferring data to and from the GPU multiple times
- Use the `alloc` clause for temporary data only needed on the device
- Use the `update` directive to synchronize data between the CPU and GPU
- Try using `firstprivate` clause for scalar data. Compiler might pass in the variable as a kernel launch argument, instead of having to map the variable to main memory

Use libraries

- Lots of available libraries for common scientific operations
- Some libraries are vendor independent (will work with AMD, NVIDIA, INTEL …) e.g. MAGMA
- Some are vendor dependent. But you can switch between implementations at build or run time, based on available hardware (e.g. rocBLAS, cuBLAS …)
- Might need `is_device_ptr`, `has_device_addr` clauses

Use asynchronous tasks

- If merging computational kernels is impossible or difficult, try to submit multiple kernels in parallel to the device
- Can use multiple threads for launching kernels. This is not always implemented efficiently in compilers.
- Use the `nowait` clause and `taskwait` construct
- Overlap computation on the device with other tasks on the CPU (e.g. communication)

Reduce OpenMP parallel overhead

- Try to always have
  `target teams distribute parallel for`
  in the same line
- (overhead of distributing teams) + (overhead of creating a parallel region in a team) > (overhead of distributing teams and creating a parallel region in one step)

Reduce OpenMP parallel overhead

```
#pragma omp target teams distribute
{
    // some code here
    #pragma omp parallel for
    {
    // some code here
    }
}
```
C/C++

- Larger overhead

```
#pragma omp target teams distribute parallel for
{
    // some code here
}
```
C/C++

- Smaller overhead

Reduce OpenMP parallel overhead

```
!$omp target teams distribute

 ! some code here
 !$omp parallel do
 ! some code here

 !$omp end target teams
```
F

- Larger overhead

```
!$omp target teams distribute parallel do
 ! all code here
 !$omp end target teams
```
F

- Smaller overhead

Conclusions

- Ensure there is enough parallelism to fill all the compute units
- Avoid branches
- Optimize memory accesses, make sure neighbouring threads access neighbouring data elements in memory
- Have kernels sufficiently big to hide latency, but sufficiently simple not to run out of registers